

Program: Diploma in Food Processing Technology	
Course Code: 5421	Course Title: Unit Operations of Food Industry
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4(L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

1. To understand the concept of evaporation and the working of different types of evaporators
2. To study the freezing mechanism in food, its principles and related equipments.
3. To impart knowledge on the effect of size reduction and filtration and its equipments

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of chemistry	Secondary school

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concepts of evaporation in food processing and different types of evaporators	17	Understanding
CO2	Describe the principles and mechanisms of freezing and freezing systems	17	Understanding
CO3	Describe the principles, working mechanisms, and applications of various food processing equipment, including hammer mills, ball mills, attrition mills, and colloidal mills, in relation to energy laws and their effects on food.	13	Understanding
CO4	Describe the theory, importance, equipment, mechanisms, and applications of filtration, including plate and frame filters, rotary filters, and the use of filter aids.	13	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concepts of evaporation in food processing and different types of evaporators		
M1.01	Discuss the Principle of evaporation	6	Understanding
M1.02	Explain the working principle and mechanism of Natural, forced circulation & falling thin film evaporator-	7	Understanding
M1.03	Describe Single effect and multiple effect evaporator	4	Understanding
Contents: Principles of evaporation, factors affecting rate of evaporation, types of evaporators, natural, forced, thin film evaporators etc. single effect, multiple effect evaporators.			
CO2	Describe the principles and mechanisms of freezing and freezing systems		
M2.01	Describe the Principle of freezing	7	Understanding
M2.02	List the factors influencing freezing time	3	Remembering
M2.03	Describe the direct freezing systems	3	Understanding
M2.04	Describe the Indirect freezing systems	3	Understanding
	Series Test – I	1	

Contents: Introduction, principles of freezing,. freezing point curve for food and water, factors influencing freezing time,Freezing systems; Direct contact systems(air blast, immersion) & indirect contact systems			
CO3	Describe the principles, working mechanisms, and applications of various food processing equipment, including hammer mills, ball mills, attrition mills, and colloidal mills, in relation to energy laws and their effects on food.		
M3.01	State Principles of size reduction and Energy laws	5	Understanding
M3.02	Discuss the equipment of size reduction in solid food	5	Understanding
M3.03	Discuss the equipment of size reduction in liquid food	3	Understanding
Contents: Principles, types of equipments, energy laws, effect on foods,- mechanism and working principles of -- hammer mill, ball mill, attrition mill, colloidal mill			
CO4	Describe the theory, importance, equipment, mechanisms, and applications of filtration, including plate and frame filters, rotary filters, and the use of filter aids.		
M4.01	Explain the theory of Filtration, importance of filtration	4	Understanding
M4.02	Discuss about the filtration equipment	4	Understanding
M4.03	Summarize filter aids	4	Understanding
	Series Test – II	1	
Contents: Theory of filtration, importance of filtration, Filtration equipment, mechanism and diagrams with applications-plate and frame, rotary filters, Filter aids-importance and applications			

Text / Reference

T/R	Book Title/Author
R1	Earle R L Unit Operations in Food Processing. Web Edition. Pergamon Press, UK2004

R2	Fellows P J, Food Processing Technology Principles and Practice, CRC Press, 2000.
R3	Geankoplis C J, Transport Process and Unit Operations, Prentice Hall Intl Edition, 1993
R4	Albert Ibarz and Gustavo V. Barbosa-Cánovas, Unit Operations in Food Engineering, CRC Press, 2003
R5	Coulson, J.M. and etal. “Coulson & Richardson’s Chemical Engineering”, 6th Edition, Vol. I & II, Butterworth – Heinman / Elsevier, 2004
R6	Foust, A.S. etal., “ Principles of Unit Operations”, 2nd Edition, John Wiley & Sons, 1999.
R7	R. Paul Singh and Dennis R. Heldman, Introduction to Food Engineering, 5th Ed. Elsevier ,2014

Online Resources

Sl.No	Website Link
1	https://www.pharmaguideline.com/2007/02/filter-aids-and-filter-medias.html
2	https://www.sciencedirect.com/science/article/abs/pii/S0032591075850042
3	https://savree.com/en/encyclopedia/ball-mill
4	https://pharmaguideline.com/2016/01/working-and-principle-of-colloidal-mill.html